

Facebook Page Content Analysis

Three-Month Performance Dashboard

Data preparation, performance analysis, and Power BI dashboard development

Item	Details
Platform	Power BI Desktop
Analysis period	Three months (May–August 2026 shown in the dashboard)
Source	Company Facebook Page performance data
Data availability	Raw company data is excluded from the public portfolio because of confidentiality requirements.

Confidentiality: The underlying company dataset is not included in this portfolio project. The repository contains only non-sensitive documentation, visual outputs, and methodology.

1. Project Overview

The objective of this project was to transform Facebook Page performance data collected over approximately three months into a structured analytical dataset and an executive-style Power BI dashboard. The analysis focuses on audience activity, content consumption, engagement, profile visits, followers, and link clicks, with a daily trend view to identify periods of higher and lower activity.

Business questions addressed

- How did reactions and profile visits change over time?
- What was the overall level of content views and viewers during the analysis period?
- How many followers and link clicks were recorded in the available data?
- Which periods showed noticeable peaks or changes in audience activity?
- How can the page's content performance be summarized in a clear management dashboard?

2. Data Preparation & Cleaning

The raw Facebook exports were organized into a consolidated dataset so that the different extracts could be analyzed consistently. The preparation process included combining the source data, reviewing columns and data types, handling missing values, and preparing numeric fields for aggregation in Power BI.

Preparation workflow

Step	What was done
1. Consolidation	Combined the relevant Facebook performance data into one unified table for analysis.
2. Structure review	Reviewed column names, date fields, and numeric performance fields to make them suitable for reporting.
3. Missing values	Reviewed and handled missing values during data preparation to prevent incomplete records from affecting the analysis.
4. Metric preparation	Prepared engagement and audience-related fields so they could be aggregated consistently in the dashboard.
5. Validation	Checked the resulting totals and trends against the source reporting context before building the final visuals.

Aggregation terminology: The current dashboard uses SUM aggregations for its displayed KPI and chart metrics. The term “average” should only be used when an actual average calculation has been applied to a field.

3. Data Model & Metric Aggregation

The Power BI report uses a consolidated table named Unified Data as the main analytical source. The displayed visuals use the following aggregation logic:

Dashboard metric	Aggregation	Purpose
Reactions	SUM	Total recorded reactions across the selected period.
Profile Visits	SUM	Total recorded profile visits across the selected period.
Followers	SUM	Displayed as the sum of the Followers field.
Link Clicks	SUM	Total recorded link clicks across the selected period.

3. Data Model & Metric Aggregation (continued)

Dashboard metric	Aggregation	Purpose
Views	SUM	Total recorded content views used in the comparison chart.
Viewers	SUM	Displayed as the sum of the Viewers field.

Metric-definition consideration: SUM is appropriate for event or count metrics when each row represents incremental activity. If Followers or Viewers are daily snapshots or running totals, summing the daily values may overstate the three-month result. The final aggregation should therefore follow the definition of each Facebook export field.

4. Dashboard Design & Visual Analysis

The dashboard is designed as a compact management overview. It combines KPI cards with two charts: a daily time-series comparison of reactions and profile visits, and a side-by-side comparison of views and viewers.

KPI cards

Metric	Displayed value
Reactions	143K
Profile Visits	135K
Followers	77K
Link Clicks	80

Chart 1 — Content Visits and Reactions

The line chart plots daily Reactions and Profile Visits across the analysis period. The trend shows relatively low activity during parts of May, followed by clear spikes in June and July. This makes the visual useful for identifying dates or content periods that deserve closer review.

Chart 2 — Content Views and Viewers

The comparison chart shows approximately 5.9M Views and 3.8M Viewers as displayed in the dashboard. The difference between the two measures can help assess the scale of content consumption, while the exact interpretation should follow the definitions of the source metrics.

4. Dashboard Design & Visual Analysis (continued)

What the dashboard communicates well

- It provides a quick executive summary without exposing the underlying company data.
- It separates overall KPI totals from daily trends, helping distinguish scale from timing.
- The spike pattern makes it easier to identify periods that can be investigated against individual posts or campaigns.
- The relatively low number of link clicks is visible against the much larger volume of views and engagement activity.

5. Analytical Findings & Recommendations

The dashboard indicates substantial content consumption and engagement during the three-month period, with the strongest activity concentrated around specific dates rather than being evenly distributed.

Key observations

Observation	Analytical implication
Large volume of Views	Content generated significant exposure and consumption during the period.
Reactions and Profile Visits show sharp spikes	Some publishing dates or content formats appear to have performed considerably better than the baseline.
Link Clicks are low relative to the other KPIs	If driving traffic is a business goal, the content-to-click journey should be investigated.
Follower total requires metric-definition validation	Confirm whether the source field represents incremental follows or a cumulative daily snapshot before using the total as a business KPI.

Recommended next steps

- Join the daily performance table with post-level content data so the spikes can be traced to specific posts.
- Break results down by content format, topic, publishing time, and campaign where available.
- Validate the definitions of Followers, Viewers, Views, and other audience metrics before choosing SUM, MAX, LAST, or another aggregation.
- Add rate-based KPIs such as engagement rate or click-through rate only when the numerator and denominator use compatible definitions.
- Add date and content-type filters to make the dashboard more interactive where appropriate.

6. Portfolio / GitHub Documentation

Because the raw dataset belongs to a company and cannot be publicly shared, the repository should demonstrate the analytical workflow without exposing confidential information. The recommended portfolio version contains the Power BI report where permitted, screenshots, documentation, and a clear confidentiality statement.

Recommended repository structure

Path	Recommended content
README.md	Project overview, objectives, tools, methodology, dashboard screenshot, key findings, and confidentiality note.
report/Facebook_Page_Content_Analysis.pbix	Power BI report, only if company policy permits sharing the PBIX file.
docs/Facebook_Page_Content_Analysis_Portfolio_Report.pdf	Detailed project documentation.
images/dashboard.png	Public dashboard screenshot with no sensitive information.
data/README.md	Explain that raw company data is intentionally excluded.

Suggested GitHub project description

Facebook Page Content Analysis — Power BI

A three-month Facebook Page performance analysis built in Power BI. The project covers data consolidation, cleaning, missing-value handling, metric aggregation, time-series analysis, and dashboard design. The underlying company dataset is not included because of confidentiality requirements.

Confidentiality statement

The analysis was completed using company-owned Facebook Page data. The raw dataset and any files containing confidential business information are intentionally excluded from the public repository. The published project contains only non-sensitive documentation, visual outputs, and methodology.

7. Project Methodology Summary

Data preparation: Consolidated the available Facebook performance exports, reviewed the structure and data types, handled missing values, and prepared the fields for analysis.

Data transformation: Standardized the analytical fields and applied aggregation logic appropriate to the dashboard metrics.

Analysis: Examined overall performance KPIs and daily trends in reactions, profile visits, views, viewers, followers, and link clicks.

Visualization: Built a Power BI dashboard combining KPI cards, a daily trend chart, and a views-versus-viewers comparison.

Business consideration: Metric definitions should be checked before interpreting cumulative audience fields such as Followers or Viewers as additive totals.

This document is intended as the public methodology and project documentation for a professional data analytics portfolio.